MILITARY SPECIFICATION

ANTIDOTE, NERVE AGENTS, INJECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers Nerve Agent Antidote in a pressure-activated automatic injector suitable for intramuscular administration (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-N-196	- Needles, Hypodermic; for Luer Syringes.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116	- Preservative-Packaging, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Physical characteristics.

3.1.1 Injector. Shall be a pressure-activated, automatic injector suitable for intramuscular administration of the contained medication. The injector shall have a length no greater than 6.0 inches and a diameter no greater than 0.780 inch. The injector face through which the needle protrudes shall have a diameter of 1/2 (+1/16 or -0) inch.

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3.1.2 Needle. The needle used in the injector shall be between 20 and 22 gauge inclusive as defined in GG-N-196 when tested as specified therein.

3.2 Medication. The medication in the injector shall contain the ingredients listed in table I.

TABLE I. Medication.

Ingredient	Amount
Atropine sulfate, USP $(C_{17}H_{23}NO_3)_2 H_2SO_4.H_2O$	1.03 mg
Benactyzine hydrochloride	4.14 mg
N,N'-Trimethylene-bis-(pyridinium-4-aldoxime) dibromide (TMB-4)	39.24 mg (on the Anhydrous basis)
Methyl paraben	1.0 mg
Propyl paraben	0.10 mg
Water for Injection, USP	Sufficient to make a total volume of 2.0 ml
Hydrochloric acid	Sufficient to adjust pH to 2.7 ± 0.1
The formulation will be on the basis of the stated active ingredients and the amounts will be 93 percent to 107 percent of the amount stated.	

3.2.1 Clarity. The medication in the injector shall be clear and free from undissolved or any particulate matter when removed from the injector and examined in a suitable colorless, glass vessel.

3.2.1.1 Color. The medication shall be colorless to no more than a slight tinge of yellow.

3.2.2 Assay. Assay of the active ingredients in the finished injection shall be determined as specified in 4.2.7.

3.2.2.1 Reference standards. Reference standards of Benactyzine Hydrochloride and TMB-4 are available upon request to contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

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3.3 Ingredients.

3.3.1 Atropine sulfate. The atropine sulfate used in the manufacture of the antidote solution (injection) shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.2 Methyl paraben. The methyl paraben used in the manufacture of the antidote solution (injection) shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.3 Propyl paraben. The propyl paraben used in the manufacture of the antidote solution (injection) shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.4 Hydrochloric acid. The hydrochloric acid used in the manufacture of the antidote solution (injection) shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.5 Water for injection. The water for injection used in the manufacture of the antidote solution (injection) shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.6 Benactyzine hydrochloride. The Benactyzine hydrochloride used in the manufacture of the antidote solution (injection) shall comply with the following:

Assay (Benactyzine Hydrochloride)	Shall be not less than 96.0 percent (on as is basis).
Chloride	Shall be not less than 9.40 percent and not more than 9.80 percent.
Amine (on an anhydrous basis)	Shall be not less than 97.0 percent and not more than 101.0 percent.
Water	Shall be not more than 1.0 percent.
Melting Point (on a dried basis)	Shall be not less than 179.0°C and not more than 182.0°C.
Chromatography, TLC	Shall contain no more than two spots; the major spot shall have the same R_f as Benactyzine HCl reference standard spotted on the same plate.

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The assay shall be performed as specified in 4.2.5.1.
The chloride determination shall be performed as specified in 4.2.5.2.
The amine determination shall be performed as specified in 4.2.5.3.
The water content shall be determined as specified in 4.2.5.4.
The melting point shall be determined as specified in 4.2.5.5
Chromatographic purity shall be determined as specified in 4.2.5.6.

3.3.7 N,N'-Trimethylene-bis-(pyridinium-4-aldoxime)dibromide (TMB-4). The N,N'-Trimethylene-bis-(pyridinium-4-aldoxime) dibromide (TMB-4) used in the manufacture of the antidote solution (injection) shall comply with the following:

Assay (N,N'-Trimethylene-bis-(pyridinium-4-aldoxime) dibromide (TMB-4))	Shall not be less than 97.0 percent (on as is basis).
Bromide	Shall be not less than 35.30 percent and not more than 36.30 percent.
Amine (on an anhydrous basis)	Shall be not less than 97.0 percent and not more than 101.0 percent.
Water	Shall be not more than 1.0 percent.
Thin Layer Chromatographic Assay	Shall have a purity of not less than 97.0 percent of the reference standard.

The assay shall be performed as specified in 4.2.6.1.
The bromide determination shall be performed as specified in 4.2.6.2.
The amine determination shall be performed as specified in 4.2.6.3.
The water content shall be determined as specified in 4.2.5.4.
The thin layer chromatographic assay shall be determined as specified in 4.2.6.4.

3.4 Activation force. The injector shall activate when a force of 2.0 to 8.0 pounds (lb) is axially applied as specified in 4.2.4.1. The front end of the injector shall not become dislodged during activation.

3.5 Volume of medication and dispensing time. The injector shall dispense 2.0 ± 0.2 milliliters (ml) of medication during a period of no more than 4 seconds from the time of activation when tested as specified in 4.2.4.2.

3.6 Needle length after medication is dispensed. After the injector is activated and the medication is dispensed, the free (exposed) length of the needle shall be 18 to 23 millimeters (mm) when tested as specified in 4.2.4.3.

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3.7 Rough handling and safety mechanism. The injector shall contain a safety mechanism capable of preventing accidental activation or dispensing of medication due to dropping, snagging, bumping, or other rough handling when tested as specified in 4.2.4.4. The injector shall not activate with the safety mechanism in place when a force of 20.00 ± 0.25 lbs is axially applied as specified in 4.2.4.4.

3.8 Ability to withstand high altitude (low pressure). When the injector is conditioned for 20 hours at an absolute pressure of no more than 196 mm of mercury and then for 4 hours at normal pressure and this conditioning is repeated three times (total conditioning time of 4 days), the injector shall show no evidence of medication leakage when tested as specified in 4.2.4.5.

3.9 Sterility. The medication and all components and surfaces in contact with the medication shall be sterile. When the injector is activated and the medication is ejected, the liquid shall be sterile when tested in accordance with the USP. After the injector has been conditioned as specified in 3.8, the medication and all components and surfaces in contact with the medication shall be sterile when tested as specified in 4.2.4.6.

3.9.1 Certificate of sterility. A certificate of sterility, listing all lot or control numbers, shall state that samples from each sterility lot have been tested in accordance with USP method and found to be sterile.

3.10 Accelerated aging. The injectors shall meet the requirements of table II after accelerated aging and testing as specified therein. The aged injectors shall be conditioned at 70° ± 5°F for 24 hours prior to testing. All plastic parts of the injector shall show no evidence of stress cracking (including hairline cracks) after accelerated aging.

TABLE II. Accelerated aging.

Accelerated aging conditions	Requirement	Test Paragraph
Not less than 24 hours at -65° ± 5°F	3.4 3.5 3.7	4.2.4.1 4.2.4.2 4.2.4.4
14 days at 130° ± 2°F	3.4 3.5	4.2.4.1 4.2.4.2

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3.11 Identification. The injector shall have the needle end permanently identified by a black color. The safety mechanism shall be permanently identified by a gray color. Each end shall be capable of being identified by sight and by touch.

3.12 Labeling-immediate containers. Each immediate container (injector) shall be durably and legibly marked or labeled as specified. When tested as specified in 4.2.4.7, the marking shall remain entirely legible; any labels used shall be unaffected and shall remain entirely adhered to the injector.

3.12.1 Label information. Each immediate container label shall bear the following information only. The label of the immediate container shall be brown and shall contain the following:

(a) Name and address of manufacturer (Black)
(b) Lot number (Black)
(c) Date of manufacture (Black)
(d) List of active ingredients in metric equivalents (Yellow on Black)
(e) National Stock Number (Yellow on Black)
(f) "For Use in Nerve Agent Poisoning Only" (Yellow on Black)
(g) Antidote, Nerve Agent (Nato Orange)
(h) Instructions (Black on Yellow)

(1) Pull out gray safety (with direction arrow to safety area).
(2) Place black end on outer thigh and push hard till injector functions (with direction arrow to needle).
(3) Hold firmly in place for 10 seconds.

(i) One stripe each of orange, brown and yellow on the needle edge (Front edge of the label).
(j) The storage statement: "STORE BETWEEN 2° - 8°C (35° - 46°F)."

3.13 Workmanship. Workmanship shall be first class throughout. The injector shall be free from visible damage such as cracks, dents, and distortions and from foreign matter such as grease, oil, and dirt. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

3.14 Delivery. Not more than 3 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.

3.14.1 Definition of date of manufacture. The date of manufacture is the date of filling or of sterilizing the final containers, or the date of manufacturer's or contractor's final quality approval of the final containers, which shall be not later than one month after the date of filling.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any Federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Quality conformance inspection.

4.2.1 Lot.

4.2.1.1 Lot for medication. For purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of the medication produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2.1.2 Lot for finished injectors. A lot shall consist of the injectors produced from one lot of the medication at one plant, under essentially the same manufacturing conditions provided the operation is continuous.

4.2.1.3 Lot for Water for Injection, USP. A lot shall not exceed that quantity which is distilled from the still(s) into one container during a 24-hour period.

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4.2.2 Sampling.

4.2.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Sampling for examination of aged injectors for stress cracking shall be conducted in accordance with inspection level S-4.

4.2.2.2 For test. The samples for each test shall be selected in accordance with MIL-STD-105, level S-4, using an AQL of 1.5 (percent defective), except for the sterility test and assay, where acceptance number shall be zero.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample injectors and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects:

TABLE III. Classification of defects (Antidote, Nerve Agents, Injector).

Categories	Defects 1/
Critical:	Marking or labeling of the injector incorrect, missing or illegible. Needle missing. Medication missing.
Major:	AQL 2.5 (percent defective)
101	Length not as specified.
102	Diameter not as specified.
103	Diameter of injector face through which needle protrudes not as specified.
104	Needle gauge not as specified.
105	Identification not as specified (3.11).
106	Workmanship (3.13).
	AQL 1.5 (percent defective)
107	Stress cracking of plastic parts after accelerated aging.

1/ Inspection is not restricted to the classified possible defects listed above.

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TABLE IV. Classification of defects (Preparation for delivery (Section 5)).

Categories	Defects 1/
Critical:	None defined.
Major:	AQL 4.0 (percent defective)
101	Polyethylene bag damaged, not sealed, or not as specified.
102	Marking for injectors not readable through bag.
103	Flexible plastic cushioning material missing or not as specified.
104	Contents per shipping container not as specified.
105	Shipping container damaged.
106	Marking of intermediate and exterior containers incorrect, missing, or illegible.

1/ Inspection is not restricted to the classified possible defects listed above.

4.2.3.2 For test. Samples selected in accordance with 4.2.2.2 shall be tested as specified in 4.2.4.

4.2.4 Tests. All injectors shall be equilibrated at a temperature of 70°F ± 5°F for at least one hour and maintained at that temperature during testing. Tests shall be conducted as follows:

4.2.4.1 Activation force. Remove the safety mechanism from the injector. Place the injector in a suitable device capable of applying and measuring required loads. Apply a load axially to the injector with the least possible impact and increase the load until the injector is activated. Record the actual load at which the injector was activated. Examine the injector to determine if the front end has become dislodged during activation.

4.2.4.2 Volume of medication and dispensing time. Using suitable apparatus, activate the injector and quantitatively collect the dispensed medication in a container suitable for volume determination by weight, corrected for specific gravity. Measure the volume of medication and to the nearest second the time between activation and the stop of the flow of medication from the needle.

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4.2.4.3 Needle length after medication is dispensed. Using suitable apparatus, activate the injector. After the medication has stopped flowing from the needle, measure the length of the needle protruding from the injector to the nearest millimeter.

4.2.4.4 Rough handling and safety mechanism. After accelerated aging as specified in the applicable portions of table II, drop the injector onto a concrete floor from a height of 4 feet. Repeat the drop three times. Examine the injector for evidence that it would be unusable. Without removing the safety mechanism, attempt to activate the injector by applying a force of 20.00 ± 0.25 lb.

4.2.4.5 Ability to withstand high altitude (low pressure). Place the injector in a suitable vacuum chamber and reduce the pressure to 196 mm of mercury absolute and maintain the pressure for 20 hours. Return the pressure to normal for 4 hours. Repeat the pressure cycle 3 times (total of 4 cycles). Remove the injector from the chamber and visually examine the injector for evidence of medication leakage.

4.2.4.6 Sterility. Condition the injector as specified in 4.2.4.5. Remove the filled cartridge and rubber needle guard component from the injector. Penetrate the rubber needle guard with the needle. Eject the medication from the cartridge and test the ejected medication for sterility in accordance with the applicable portions of the sterility tests contained in the USP.

4.2.4.7 Marking. Rub the marking or labeling on the injector ten times with fingers using moderate pressure. Examine for legibility and permanency of marking or labeling.

4.2.5 Tests - Benactyzine hydrochloride.

4.2.5.1 Benactyzine hydrochloride. (Note: A general discussion on gas chromatography is included in the USP starting on page 639. It is suggested that this information be reviewed as background for apparatus, columns and system suitability tests.)

Reagents

Chloroform, ACS grade

Sodium bicarbonate solution - 7 gram/100 ml of distilled water. Must be freshly prepared.

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Chromatographic gases - Helium, oil-free; air, filtered through molecular sieve 5X after oil trap; hydrogen, oil-free.

Benactyzine hydrochloride reference standard

Apparatus

Gas chromatograph equipped with flame ionization detector, glass column, and a means for integrating the peak area.

Column packing - 3 percent OV-17 on 100/120 mesh Gas chrom-Q, available from Applied Science Labs, State College, PA 16801.

Chromatographic parameters

Column	glass, 4mm ID X 6 ft length
Column support & coating	100/120 mesh Gas chrom-Q with OV-17 (3 percent loading)
Column temperature	200°C isothermal
Injection port	190°C
Detector temperature	220°C
Carrier gas & flow rate	Helium, 80 ml/min.
Detector fuel	Hydrogen, 80 ml/min.
Detector air	450 ml/min.

Retention Time

The retention time of benactyzine is approximately 7.0 minutes.

Procedure

Standard Curve Preparation

Benactyzine hydrochloride. Weigh a 200 mg sample of standard to the nearest 0.1 mg of benactyzine hydrochloride into a 10 ml volumetric flask. Dilute to volume with distilled water.

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Procedure. With a pipet remove a 1.0 ml aliquot of the above standard solution to a clean 5 ml screw capped vial with a teflon liner. Add 1 ml of a 7 percent aqueous sodium bicarbonate solution which is prepared daily. With a pipet, add 1.0 ml of ACS grade chloroform to the above aqueous solution and shake vigorously for 2 min. Allow the two layers to separate completely. By use of a micro syringe, remove an aliquot from the bottom (chloroform) layer and inject separate 3, 4, 5 and 6 ul of the chloroform layer into the gas chromatograph. The area response for each benactyzine quantity injected is plotted to form a standard curve.

Sample Procedure

For assay, weigh a 200 mg sample of the benactyzine hydrochloride, to the nearest 0.1 mg, into a 10 ml volumetric flask and dilute to volume with distilled water. Pipet a 1.0 ml aliquot of this solution and treat as in the above procedure for the calibration curve. Inject a measured 3 to 6 ul aliquot of the chloroform layer into the gas chromatograph.

Calculations. Compare the peak areas obtained from the injected sample with the standard curve to obtain the amount of benactyzine injected as the hydrochloride.

$$\text{Percent benactyzine hydrochloride} = \frac{A}{B \times W}$$

Where:

A = micrograms of benactyzine as the hydrochloride from the standard curve

B = microliters of chloroform injected

W = Weight of sample in grams

4.2.5.2 Chloride. The chloride content shall be determined by the USP procedure for a chloride as in the assay of Potassium Chloride, USP, suitably modified to remove the silver chloride precipitate and revising the calculation to reflect the proper milli-equivalent.

4.2.5.3 Amine. The amine content shall be determined as follows:

When titrating, the light source must be incandescent and not fluorescent light.

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Reagents

Acetic acid solvent. Prepare a solution of 20 ml of acetic anhydride per liter of glacial acetic acid. Age 24 hrs before use.

Crystal violet indicator, 0.1 percent in glacial acetic acid.

Perchloric acid, 0.1N, dissolve 8.5 ml of 70 percent perchloric acid in one liter of glacial acetic acid containing 20 ml of acetic anhydride and age for 24 hrs. Standardize the reagent as follows: Weigh approximately 0.15 gram of NBS potassium acid phthalate to the nearest 0.1 mg and dissolve in 50 ml of acetic acid solvent (previously blanked out with 0.1N perchloric acid using 4 to 5 drops of crystal violet indicator) in a dry 125 ml Erlenmeyer flask. Titrate with the perchloric acid solution to a complete color change from purple to blue. Calculate the normality of the perchloric acid as follows:

$$\text{Normality} = \frac{W}{0.2040\ A}$$

Where: W = grams of NBS potassium phthalate

A = milliliters of perchloric acid

Procedure for Amine Titration

Weigh 250 to 350 mg of sample to nearest 0.1 mg into an oven dried 125 ml Erlenmeyer flask and add 25 ml of acetic acid solvent (previously blanked out as described in perchloric acid standardization). Dissolve the sample by swirling and titrate with the standardized perchloric acid to a complete color change from purple to blue. Near the end point add mercuric acetate in 10 mg amounts and continue the titration until the addition of mercuric acetate does not cause the purple color to reappear. Calculate the percent amine as follows:

$$\text{Percent Amine} = \frac{A \times B \times 36.39}{W}$$

Where: A = milliliters of perchloric acid
B = normality of perchloric acid
W = weight of sample in grams

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The percent amine on an anhydrous (water free) basis is calculated by the following equation.

$$\text{Percent Amine (water free)} = \frac{A \times 100}{B}$$

Where: A = percent amine on sample as received
B = 100 percent water in sample

4.2.5.4 Water. The water content of the material as received shall be determined by method 1 of the USP using an accurately weighed sample between 500 to 1000 mg.

4.2.5.5 Melting Point. The melting point of the material shall be determined by the USP method as defined for class 1a, using a sample previously dried to constant weight at 120°C (about 3 hours).

4.2.5.6 Chromatography, TLC. Chromatographic purity will be determined as follows.

Thin Layer Chromatographic Plates

Silica gel with fluorescent indicator (254 mu), 0.25 mm thickness on glass.

Developing Solvent.

Methanol:n-butanol:glacial acetic acid:water (40:40:5:5).

Benactyzine Hydrochloride solution.

Weigh accurately 20 mg of benactyzine hydrochloride; dissolve in 10.0 ml 50 percent ethanol solution.

Benactyzine Hydrochloride Reference Standard Solution.

Weigh accurately 20 mg benactyzine hydrochloride reference standard; dissolve in 10.0 ml of 50 percent ethanol solution.

50 percent Ethanol Solution.

To 5.0 ml absolute ethanol add sufficient purified water to make 10.0 ml of solution.

Procedure.

Activate chromatographic plates by heating at 90°C for 30 minutes, cool. Apply 50 ul of benactyzine solution to 4 spots, 20 mm from the bottom of the plate, 20 mm apart. On the same plate apply 50 ul of a benactyzine reference standard solution to 4 spots 20 mm from the bottom of the plate, 20 mm apart. To a separate spot 20 mm from any other spot and 20 mm from the bottom of the plate apply 50 ul of the 50 percent ethanol solution for the reference blank.

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Develop plates at room temperature in a closed container with the developing solvent to achieve a solvent front 170 mm from the bottom of the plate. Remove the plates from the container and allow to dry in air at room temperature until solvent is no longer detectable.

Observe fluorescent quenching by viewing the plates in darkness, illuminated with short wave ultraviolet light (254 mu). Compounds will appear as darkened spots on a fluorescent background. The R_f for benactyzine is 0.42.

4.2.6 Tests-TMB-4.

4.2.6.1 TMB-4.

Accurately weigh to the nearest 0.1 mg approximately 100 mg of sample into a 1 liter flask dissolved in, and diluted to volume with purified water USP. Pipet 5.0 ml into a 100-ml volumetric flask and dilute to volume with 0.1N NaOH solution.

Similarly prepare a reference standardwith a known concentration of approximately 5 mcg per ml. Determine the absorbances of sample and standard in 1-cm cells at the wave length of maximum absorbance, at approximately 346 nm, with a suitable spectrophotometer, using 0.1N NaOH solution as the blank.

Calculation:

$$\frac{Au}{AS} \times \text{exact concentration standard mg/ml} \times \frac{1000}{\text{sample wt (mg)}} \times \frac{100}{5} \times 100 = \text{percentage}$$

Where:
Au = Absorbance sample
As = Absorbance standard

4.2.6.2 Bromide. The bromide content shall be determined by the NF procedure as in the "bromide content" for Mepenzolate Bromide suitably modified by revising the calculation to reflect the proper milliequivalent.

4.2.6.3 Amine. The amine content shall be determined as follows:

When titrating, the light source must be incandescent and not fluorescent light.

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Reagents

Acetic acid solvent. Prepare a solution of 20 ml of acetic anhydride per liter of glacial acetic acid. Age 24 hrs before use.

Crystal violet indicator, 0.1 percent in glacial acetic acid.

Perchloric acid, 0.1N, dissolve 8.5 ml of 70 percent perchloric acid in one liter of glacial acetic acid containing 20 ml of acetic anhydride and age for 24 hrs. Standardize the reagent as follows: Weigh approximately 0.15 gram of NBS potassium acid phthalate to the nearest 0.1 mg and dissolve in 50 ml of acetic acid solvent (previously blanked out with 0.1N perchloric acid using 4 to 5 drops of crystal violet indicator) in a dry 125 ml Erlenmeyer flask. Titrate with the perchloric acid solution to a complete color change from purple to blue. Calculate the normality of the perchloric acid as follows:

$$\text{Normality} = \frac{W}{0.2040\ A}$$

Where: W = grams of NBS potassium phthalate

A = milliliters of perchloric acid

Procedure

Weigh 100 to 250 mg of sample to the nearest 0.1 mg into a 125 ml Erlenmeyer flask, being careful to get all the sample in a small area of flask. With a pipet add 1 ml of water and swirl gently to dissolve sample. Pipet 20 ml of acetic anhydride, cover flask with a small watchglass and bring to a boil on a hot plate. Allow sample to cool to room temperature with occasional swirling. Titrate with 0.1N standardized perchloric acid to a complete color change from purple to blue. Near the end point add mercuric acetate in 10 mg amounts and continue the titration until the addition of mercuric acetate does not cause the purple color to reappear. A blank is run parallel with the sample.

Calculation:

$$\text{Percent Amine} = \frac{223.1 \times B \times (A-C) \times 100}{W \times 1000}$$

B = normality of perchloric acid

A = milliliters of standard perchloric acid for sample titration

C = milliliters of standard perchloric acid for blank

W = sample weight in grams

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The percent amine on an anhydrous (water free) basis, is calculated by the following equation.

$$\text{Percent amine (water free)} = \frac{A \times 100}{B}$$

Where: A = percent total amine on sample as received

B = 100-percent water in sample

4.2.6.4 Thin layer chromatographic assay of TMB-4. Dissolve a suitable quantity of TMB-4, accurately weighed, in dilute hydrochloric acid (1 in 5000) and dilute quantitatively with the dilute acid to obtain a solution having a known concentration of about 12 mg per ml. Dissolve an accurately weighed quantity of TMB-4 Reference Standard in dilute hydrochloric acid (1 in 5000) and dilute quantitatively with the dilute acid to obtain a Standard Solution having a known concentration of about 12 mg per ml. Chromatograph concomitantly 5 samples of each of the above solutions and one sample of the dilute acid, the later for use as a blank.

Procedure. On a suitable thin layer chromatographic plate coated with a 250-mm layer of Q5F silica gel (Quantum Industries, or its equivalent), apply accurately 10 microliters of the solutions to form a narrow streak about 30 mm across the plate. Allow the spot to dry, and develop the chromatogram in a solvent system consisting of n-butanol, acetic acid and water (5:1:2). After the solvent front has moved at least 10 cm from the origin, remove the plate from the developing chamber and allow the solvent to evaporate. Locate the TMB-4 spot by visualization under 254 nm untraviolet light. The TMB-4 will be the major spot found in the chromatogram of the standard solution. Typically of the Rf of TMB-4 should be between 0.1 and 0.2. Thoroughly scrape identical areas of the plates and collect the absorbate from each plate in suitable containers. The scraping area chosen must be such as to encompass the TMB-4 spot on each plate without disturbing any other spots which may be evident.

Add 15 ml of 0.05N NaOH to each sample and shake the mixture vigorously for 15 minutes. Filter the mixture through a 0.45-micrometer Millipore filter, discarding the first 2 ml of filtrate. Determine the absorbancies of the filtered solutions in 1-cm cells at the wavelength of maximum absorbance at about 346 nm, with a suitable spectrophotometer, using the filtrate from the blank plate as the blank.

Determine the average absorbancies obtained for five (5) TMB-4 solutions and for the five (5) standard solutions. If the absorbance of one or more samples in either group differs from the average of that group by more than 0.015 absorbance units, the assay must be repeated.

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Calculate the Relative Purity of the TMB-4 by the formula:

$$100\ (AuCs/CuAs)$$

in which Au is the average absorbance obtained from the five TMB-4 solutions. As is the average absorbance obtained from the five Standard Solutions, Cu is the concentration in mg per ml of the TMB-4 solution, and Cs is the concentration in mg per ml of the Standard Solution.

4.2.7 Assays for Final Product.

4.2.7.1 Atropine Sulfate And Benactyzine Hydrochloride.

The final product shall be analyzed for atropine sulfate and benactyzine hydrochloride by the method described for Atropine Sulfate Injection, USP.

Standard preparation of benactyzine hydrochloride will be made with reference standard provided by the Government.

Potency for the atropine sulfate and benactyzine hydrochloride in the final product should be within the limits specified for Atropine Sulfate Injection, USP.

4.2.7.2 TMB-4 Bromide. The potency of TMB-4 bromide in the Nerve Agent Antidote shall be determined as follows.

Standard Preparation.

Weigh accurately 19.62 mg of TMB-4 Reference Standard (on anhydrous basis). Transfer to stoppered siliconized glass container. Dissolve in 1.0 ml purified water. Accurately transfer 10.0 ul of this solution to a 50 ml siliconized glass volumetric flask. Dilute to 50 ml with 0.1N potassium hydroxide.

Assay Preparation.

Accurately transfer 10.0 ul of Nerve Agent Antidote Injection using a siliconized glass microsyringe or pipet to a 50 ml siliconized glass volumetric flask. Dilute to 50 ml with 0.1N potassium hydroxide.

Procedure.

Determine UV absorbance of Nerve Agent Antidote Injection and Reference Standard as specified in 4.2.6.1.

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Calculation.

$$\frac{Au}{As} \times 100 = \text{Percentage of labeled amount.}$$

4.2.7.3 Methyl Paraben and Propyl Paraben. The concentration of methyl and propyl parabens will be determined as follows.

Internal Standard Solution. Dissolve 25.0 mg of USP Butyl Paraben Reference Standard in ethyl acetate contained in a 50 ml volumetric flask, add ethyl acetate to volume and mix. Prepare fresh daily.

Standard Preparation.

Dissolve 25.0 mg of USP Methyl Paraben Reference Standard and 2.5 mg of USP Propyl Paraben Reference Standard in water contained in a 50 ml volumetric flask, add water to volume and mix. Prepare fresh daily.

Assay Preparation.

Deliver 1.0 ml of Nerve Agent Antidote Injection into a stoppered glass conical centrifuge tube (15 ml vol). Add 1 ml of 0.01N Hydrochloric Acid, 5 ml of ethyl acetate and 0.5 ml of Internal Standard solution. Mix vigorously for 3 minutes. Transfer ethyl acetate layer to a 15 ml glass conical centrifuge tube. Measure and record the volume. Filter and rinse through about 2 gram of anhydrous sodium sulfate suitably supported in a funnel. Collect the filtrate in a conical glass centrifuge tube and evaporate under gentle dry air stream to dryness. Rinse sides of tube with ethyl acetate and evaporate to dryness.

Bis-Trimethylsilylacetamide (BSA) Solution.

Dilute 1 ml of bis-trimethylsilylacetamide to 100 ml with ethyl acetate.

Procedure. To the centrifuge tubes containing the dried assay preparations, add 100 ul of bis-trimethylsilylacetamide solution. Heat the tubes for 10 minutes at 30° C, cover, cool to room temperature.

Inject 2 ul of this solution into the gas chromatograph to obtain suitable chromatograms for the extracts.

Standard Curve.

Pipet into separate stoppered glass conical centrifuge tubes 0.25, 0.50, 1.0 and 1.50 ml of the Standard Preparation and treat each solution as directed under Assay preparation.

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Apparatus.

The gas chromatographic instrument as described in 4.2.5.1 will be used for the analyses.

Chromatographic Parameters

Column	Glass, 4mm ID x 1.2 m length
Column Temperature	118° C isothermal
Injection Port Temperature	180° C
Detector Temperature	260° C
Carrier Gas and Flow Rate	Helium, 35 ml/min
Detector Fuel and Flow Rate	Hydrogen, 35 ml/min

Retention Indices 1/

Methyl paraben	1620
Propyl paraben	1783
Butyl paraben	1883
p-Hydroxybenzoic Acid	1717
Benzophenone	1831
Benzilic Acid	2003
4-Pyridin aldoxime	1467

NOTE: All substances except the parabens may occur as impurities or breakdown products of ingredients in the Nerve Agent Antidote Injection.

Calculations.

Measure the heights, H_m, H_p, and H_B of the methyl paraben, propyl paraben and butylparaben peaks respectively in each chromogram and calculate the ration, R by H_M/H_B and H_p/H_B. Plot the standard curve of the values of R for the Standard Preparation in mg of USP Methyl Paraben Reference Standard or USP Propyl Paraben Reference Standard contained in each portion of the Standard Preparation taken.

1/ References: Kobats Retention Index, Helv. Chim. Acta Vol. 41 Page 1915, 1958; Dean, John A. "Chemical Separation Methods", 1969 Pages 235-236 Van Nostrand Reinhold, New York, NY.

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Measure the height of the paraben peaks obtained from Nerve Agent Antidote Injection and calculate the R of the height of the methyl paraben and propyl paraben peaks as ratios of the butyl paraben peaks. Read the Standard Curve quantity, in mg per ml, of the respective paraben in the volume of the Nerve Agent Antidote Injection.

5. PACKAGING

5.1 Packaging. Injectors shall be packaged level A.

5.1.1 Level A. Each injector shall be individually packaged in a clear polyethylene bag in accordance with method IC-3 of MIL-P-116. At the time of packaging each injector shall be clean and dry when tested as specified in MIL-P-116. The bag shall be formed from film of 0.003 inch minimum nominal thickness and shall have minimum inside dimensions of 7-1/4 inches by 1-1/2 inches. Each bag shall be designed to permit fast and easy opening. Two injectors, individually packaged in bags as specified above, shall be placed with points down in a pouch formed from nominal 1/8 inch thick polyethylene foam conforming to type II, class 1 of PPP-C-1752. The pouch shall have inside dimensions of 6-1/4 $\pm$ 1/8 inch by 2-3/8 $\pm$ 1/8 inch and the opening shall be in one end along the smaller dimension.

5.1.1.1 Intermediate packaging. Fifty pouches of two injectors each, packaged as specified in 5.1.1, shall be intermediately packaged with openings in the same direction in a fiberboard box conforming to grade W5c of PPP-B-636. The box shall have inside dimensions of 12-5/8 inches by 11-1/2 inches by 7-1/4 inches. Layers of pouches shall be separated by fiberboard pads formed from material conforming to minimum grade W5c of PPP-F-320. Motion of contents within the box shall be prevented by the use of cushioning material conforming to type II, class A of PPP-C-843. The box shall be closed as specified in the appendix of PPP-B-636.

5.2 Packing. Injectors shall be packed level A or C (see 6.1).

5.2.1 Level A. Two intermediate packages of injectors prepared as specified in 5.1.1 shall be packed in a fiberboard box conforming to grade V3c of PPP-B-636 for a type 2 load of 50 to 100 pounds. The box shall have inside dimensions of 23-7/8 inches by 13-1/4 inches by 8-3/4 inches. Motion of contents shall be prevented by the insertion, where needed, of fiberboard pads formed from material conforming to grade W5c of PPP-F-320. The box shall be closed and reinforced by filament-reinforced tape banding as specified in the appendix of PPP-B-636.

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5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Marking.

5.3.1 Unit package. The exterior of each polyethylene bag shall be marked with opening instructions. Markings on the bag shall not become obliterated or indistinct in handling.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Marking shall include the legend: "STORE BETWEEN 2° - 8°C (35° - 46°F)."

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Lot (control) number shall be shown. Marking shall include the legend: "STORE BETWEEN 2° - 8°C (35° - 46°F)."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.1 and 5.2).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-134-2943	ANTIDOTE, NERVE AGENTS, Injector

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6505-1398

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-37251	ANTIDOTE, NERVE AGENTS, INJECTOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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